Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

MB507 – MEDICAL BACTERIOLOGY

Time : 3 hrs **Max Marks :75**

SECTION – A (10X1=10)

Answer **ALL** the Questions

Choose the correct answer

1. Which one of the following is an obligate intracellular pathogen
   a) *Yersinia*
   b) *Chlamydia*
   c) *E. coli*
   d) *Vibrio*

2. The type of infection caused by fatal strains of a group of *Streptococcus* in the deeper layers of tissues is
   a) Scarlet fever
   b) Rheumatic fever
   c) Necrotizing fasciitis
   d) Haemolysis

3. Hemolytic-uremic syndrome
   a) affects children
   b) acute kidney failure is resulted
   c) caused by *Pseudomonas*
   d) both a and b

4. Non fluorescent bluish pigment produced by *Pseudomonas* that could aid in identification is
   a) pyocyanin
   b) pyroverdin
   c) pyorubin
   d) pyomelanin

5. The improper cooking of rice leads to food poisoning caused by
   a) B. anthracis
   b) *B. cereus*
   c) Bordetella pertussis
   d) B. subtilis

6. Pseudomembranous colitis is associated with
   a) Mycobacterium
   b) Clostridium difficile
   c) *E. coli*
   d) *Haemophilus*

7. Which of the following is NOT a recognized virulence factor of *Bordetella pertussis*?
   a) Heat labile toxin
   b) Filamentous hemagglutinin
   c) Tracheal cytotoxin
   d) pertussis toxin

8. Humans usually infected with *Legionella pneumophila* by
   a) fomites associated with legionella carrier
   b) Breathing aerosols from environmental water sources
   c) biting with carrier mosquitoes
   d) consuming undercooked meats

9. A rapid diagnostic technique used in the laboratory for confirmation of *Mycoplasma pneumoniae* infection is.
   a) culture in broth containing serum, glucose and penicillin (to inhibit other bacteria)
   b) PCR
   c) Electron microscopy
   d) EIA test on acute and convalescent phase sera

10. The histopathological hallmark of infection caused by *Rickettsia rickettsiae* is
   a) perivascular lymphocytes infiltration
   b) morulae within monocytes
   c) granulomatous inflammation
   d) intracellular vacuoles formation

SECTION – B (5X3=15)

Answer any **FIVE** Questions

11. Write short notes on endotoxin.
12. Explain the Pathogenicity of *Klebsiella*.
13. Write short notes on selective media and enrichment culturing of *Salmonella*.
14. What are the laboratory diagnostic tests performed to identify Anthrax pathogen?
15. Substantiate the statement “Protective vaccination against *Bordetella pertussis*”.
16. Critically comment on epidemiology of Q fever.

17. State the importance of Ziehl-Neelsen staining for *Mycobacterium* identification.

SECTION – C (5X10=50)

Answer **ALL** the Questions

18. a) Explain how toxins play an important role in infection.

(Or)

b) State and explain any five virulence factor associated with pathogenic bacteria.

19. a) Describe the role of antigenic structure and toxin production in *Shigella.*

(Or)

b) Explain the clinical diseases induced by *Salmonellae*.

20. a) Describe the clinical manifestation of *Bacillus anthracis.*

(Or)

b) Interpret the role of toxins in pathogenesis of *Clostridium botulinum.*

21. a) Summarize about the culture and growth characteristics. diagnostic laboratory tests to Identify *Legionella pneumophila.*

(Or)

b) Describe the antigenic structure and pathogenesis of whooping cough pathogen.

22. a) State and explain the various rickettsial fevers.

(Or)

b) How chlamydiae are classified based on pathogenic potential, host range, antigenic differences.

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